AiO-Meme: An All-in-One Multi-Agent Pipeline for Multifaceted Meme Understanding

Anonymous ACL submission

Abstract

Memes are a widely used multimodal form of communication on social media. Although Computational Meme Understanding (CMU) is inherently multifaceted, existing work remains fragmented, with disparate task definitions and isolated subtask formulations. Meanwhile, multimodal figurative language in memes, such as metaphor, sarcasm, and irony, has attracted growing attention, yet its role in downstream meme understanding remains underexplored. To address these gaps, we introduce AiO-Meme, a unified three-stage framework grounded in figurative language understanding for comprehensive meme analysis. AiO-Meme integrates (i) zero-shot multifaceted CMU via a collaborative multi-agent system (AiO-MemeAgents), (ii) Human-AI Annotation to curate a high-quality and unified dataset (Multifacet-Meme), and (iii) Agentic Distillation that transfers multi-agent reasoning into efficient student models (AiO-LLaMAs). Experiments demonstrate strong effectiveness and generalizability: AiO-MemeAgents achieves average scores of 71.16% on HatReD and 74.59% on MET-Meme, outperforming few-shot chain-of-thought prompting by up to 13.2 points. Our distilled AiO-LLaMAs workflow retains competitive performance while substantially reducing computational cost. Our code and data are available at https://anonymous.4open.science/r/AiO-Meme.

1 Introduction

Memes are a pervasive form of multimodal communication online (Pramanick et al., 2021; Lin et al., 2024b), combining images and text to convey humor, commentary, and sometimes, harmful messages. Despite extensive research in Computational Meme Understanding (CMU) (Nguyen and Ng, 2024), existing work has focused on specific aspects of meme understanding, such as hateful content detection, sentiment classification, or figurative language recognition, often under different task definitions and datasets. Thus, a unified perspective on multi-task meme understanding remains limited, motivating holistic modeling approaches.

A key direction in multifaceted CMU research is the development of multi-task frameworks that address multiple CMU tasks jointly (Chauhan et al., 2020; Zheng et al., 2025; Xu et al., 2022; Maity et al., 2022; Sharma et al., 2023; Zhong and Baghel, 2024). Nevertheless, existing approaches differ in their task coverage, with each work addressing a subset of CMU tasks rather than a comprehensive formulation (Table 1). For example, MET-Meme (Xu et al., 2022) focuses on sentiment, intensity, and intention in general memes, emphasizing metaphoric language, whereas HatReD (Kiela et al., 2020) targets social category, target, and attack type in hateful memes.

In this paper, we review the CMU literature and identify eight core CMU tasks (Table 2), spanning figurative, emotional, and social-harm dimensions. We then propose **AiO-Meme**, an *all-in-one framework that unifies these tasks* within a single multi-agent pipeline, **AiO-MemeAgents**. These tasks are not intended to constitute an exhaustive taxonomy, but rather a representative set of core tasks that recur across prior benchmarks and collectively capture key dimensions of meme understanding. Beyond integrating previously disjoint research threads, our approach gives a structured methodology for linking figurative language interpretation to downstream meme analysis. Designed with Human–AI collaboration in mind, **AiO-MemeAgents** decomposes meme understanding into interpretable agent roles, enabling structured reasoning that can be inspected, guided, and distilled.

We organize the eight distinct CMU tasks into three task groups, each addressed by **AiO-MemeAgents**, which comprises four core components: a Figurative Agent, a Router, an Experts Group, and an Orchestrator Agent. To over-

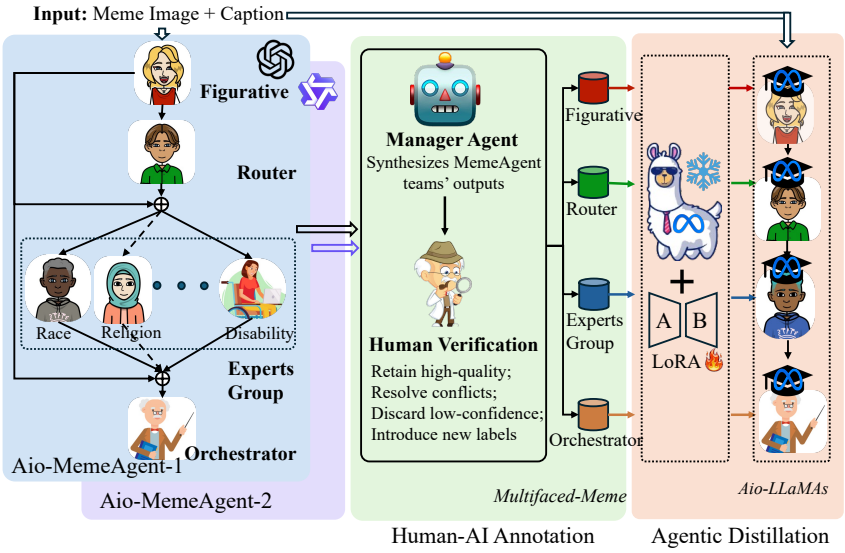


Figure 1: Overview of our proposed 3-stage pipeline **AiO-Meme**. 1) Zero-shot Multifaceted CMU uses a multi-agent framework AiO-MemeAgents; 2) Human-AI Collaboration for Annotation to curate a high-quality dataset Multifacet-Meme (Section 3.2); 3) Agentic Distillation transfers the reasoning patterns of agentic collaboration into the AiO-LLaMAs workflow consisting of small-scale fine-tuned student models (Section 3.3).

Dataset and/or Publication	#Task-Sample Pairs	Objective	#Sample	Figurative	Sentiment	Intention	Intensity	Social Category	Social Target	Attack Type	Explanations
WOAH-5* (Mathias et al., 2021)	20,000	Hate	10,000					✓		✓	
MET-Meme* (Xu et al., 2022)	40,180	General	10,045	Metaphor	✓	✓	✓				
FigMeme (Liu et al., 2022)	5,141	General	5,141	Multi-class							
HatReD* (Hee et al., 2023)	12,912	Hate	3,228					✓	✓	✓	✓
MemeIntent (Park et al., 2024)	1,900	General	950			✓					✓
Multifacet-Meme (Ours)	26,400	Mix	3,300	Multi-class	✓	✓	✓	✓	✓	✓	✓

Table 1: Dataset statistics and comparison of prior multi-task CMU studies. We aim to provide a unified work for multifaceted meme understanding. * indicates the source datasets used in this study.

come the lack of a dataset covering all eight tasks, we introduce a Manager Agent, powered by GPT-4o-mini (OpenAI, 2024), to automatically generate high-quality annotations (Subramaniam et al., 2025), supporting scalable Human–AI collaborative dataset construction. The agent synthesizes outputs from two agent teams along with background knowledge from existing datasets. Finally, we distill the reasoning capabilities of the full agentic framework into smaller student models via efficient instruction tuning (Huang et al., 2023) with LoRA (Hu et al., 2021). Our main contributions are as follows:

- We introduce **AiO-Meme**, a unified framework for multifaceted computational meme understanding spanning figurative, emotional, and harmful dimensions.
- We design **AiO-MemeAgents**, a collaborative multi-agent system for zero-shot multi-task meme understanding.
- We curate **Multifacet-Meme**, an Human–AI annotated dataset with unified labels for eight meme tasks.
- We propose **AiO-LLaMAs**, a distillation workflow that transfers multi-agent reasoning into efficient 7–13B MLLMs.
- We demonstrate strong effectiveness and generalizability across two benchmarks and five MLLM backbones. In zero-shot settings, AiO-MemeAgents outperforms few-shot chain-of-thought prompting by up to 13.2 F1 points.

2 Related work

Multi-task CMU Nguyen and Ng (2024) argue that CMU encompasses a wide range of tasks, including figurative language interpretation and fine-grained emotion recognition. A large body of work focuses on harmful meme classification (Gupta et al., 2021; Cao et al., 2022; Bianchi et al., 2022; Zhang et al., 2023a; Lin et al., 2023; Wen et al., 2023; Ocampo et al., 2023; Lin et al., 2024b), while another line of research explores multi-task CMU. For example, Sharma et al. (2020) introduce a multi-task challenge covering sentiment classification, emotion type recognition, and intensity prediction. Although several datasets support multi-

Task	Name	Type	Option list
T1	Figurative	C	metaphor, irony, hyperbole, onomatopoeia, personification, allusion, idiom, pun
T2	Social Category	C	gender, race, age, religion, disability, nationality, none
T3	Social Target	G	-
T4	Sentiment	C	happiness, love, anger, sorrow, fear, hate, surprise, neutral
T5	Intention	C	interactive, expressive, purely entertaining, offensive, other
T6	Intensity	C	Integer scores range from 1 (very mild) to 4 (intense)
T7	Attack Type	C	dehumanizing, inferiority, inciting violence, mocking, contempt, slurs, exclusion, none
T8	Explanation	G	-

Table 2: Definition of Multifaceted CMU, unifying 8 tasks from prior work (Table 1). C denotes classification tasks and G denotes generation tasks. Sentiment is the sentiment classification task defined by Xu et al. (2022).

task CMU (Mathias et al., 2021; Xu et al., 2022; Shah et al., 2024), differences in task definitions and coverage complicate cross-task integration and limit comprehensive multi-task analysis

Figurative language (e.g., metaphor and irony) plays a central role in making memes engaging and widely shared (Paul, 1970; Yosef et al., 2023; Sharma et al., 2023; Zhong and Baghel, 2024; Liu et al., 2022; Hee and Lee, 2025). Prior work shows that metaphor comprehension can enhance emotion recognition (Zhang et al., 2024). However, visual metaphors remain challenging for multimodal large language models (MLLMs) (Hwang and Shwartz, 2023), though reasoning strategies such as Chain-of-Thought (CoT) prompting (Wei et al., 2023) can improve metaphor detection (Xu et al., 2024). Other figurative phenomena, such as sarcasm and irony, have also been studied (Lin et al., 2024a; Chauhan et al., 2020), and several datasets explicitly target figurative language in memes (Yosef et al., 2023; Liu et al., 2022). Despite these efforts, the role of multimodal figurative language understanding in enabling comprehensive CMU remains underexplored.

There have been some efforts toward developing all-in-one frameworks for multi-task CMU. Early work includes joint modeling of multiple meme attributes, such as humor, sarcasm, offensiveness, motivation, and sentiment (Chauhan et al., 2020). Subsequent approaches are dataset-specific and based on relatively small foundation models, including BERT-based text encoders with VGG or ResNet vision backbones (Maity et al., 2022; Shah et al., 2024). In this work, we build on these efforts and propose an all-in-one multi-agent framework for multi-task CMU.

Knowledge Distillation Knowledge distillation (Hinton et al., 2015) is a standard approach for transferring knowledge from large teacher models to compact student models. Recent studies extend this paradigm by using LLMs and multi-agent systems as teachers that provide rich reasoning traces for training efficient models (Ravi et al., 2024; Zhao et al., 2024; Li et al., 2024; Chen et al., 2024). Ng (2025) formalize this setting as *agentic distillation*. To our knowledge, this paradigm has not been explored for multifaceted computational meme understanding, where both reasoning quality and deployability are critical.

3 Method

We describe **AiO-Meme** in this section, a three-stage pipeline grounded in multi-agent collaboration and figurative language understanding for unified multifaceted CMU. We hypothesize that multimodal figurative reasoning contributes to downstream sub-tasks, a claim supported by the ablation results shown in Figure 4.

3.1 A Multi-Agent Framework for CMU

Let $D = \{x_1, x_2, \ldots, x_n\}$ be a dataset of n memes, where each meme $x \in D$ has an image x_v and caption x_t. The goal of **AiO-MemeAgents** is to perform zero-shot, multi-task CMU on this input. While Huang et al. (2024) introduced an agent-based framework for harmful meme classification, it targets a specific CMU setting and does not consider figurative language or broader multifaceted modeling. We extend this line of work by proposing a novel agentic framework for Multifaceted CMU that supports both harmful and general memes. We decompose the problem into 4 task layers and design the AiO-MemeAgents architecture according to the task-level Mixture-of-Experts (MoE) principle (Jiang et al., 2024; DeepSeek-AI et al., 2024). The roles and design of the four task-specific agents are described below.

Figurative Agent We design a Figurative Agent,

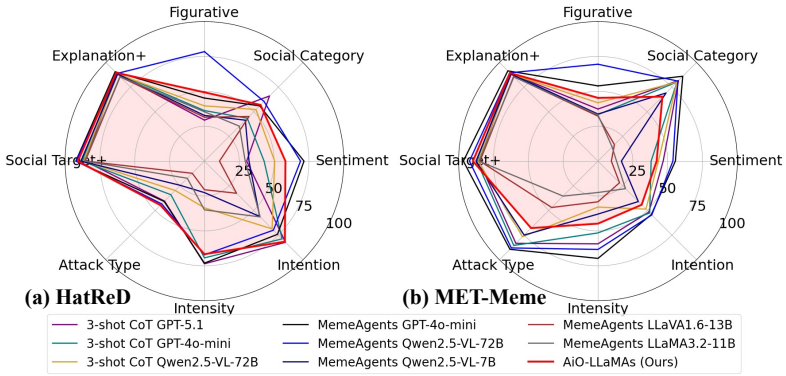


Figure 2: Performance comparison by task between zero-shot AiO-MemeAgents with different backbones and the distilled AiO-LLaMAs. Detailed results are provided in Table 3.

denoted as A_f, to serve as a shared expert for task T1: Figurative Language Understanding. We hypothesize that figurative expressions play a key role in uncovering the deep implicit semantics embedded in memes and can significantly contribute to downstream tasks T2-T8. Let p_f represent the prompt for the Figurative Agent (see Appendix A for the prompt details), and let m be the backbone MLLM used by agent. The output generated by the agent is denoted as y_f,

$$y_f = A_f(x_v, x_t, p_f, m) \tag{1}$$

Router We design a Router A_r with prompt p_r for T2, Social Category Classification (Hee et al., 2023), based on the analysis from Figurative Agent y_f. Let $\mathcal{C}$ = {gender, race, age, religion, disability, nationality} be the set of social categories. The Router determines if a social category $s \in \mathcal{C}$ is potentially involved in or affected by the meme. The decision will help to narrow down the social dimensions relevant to the meme's content and activate expert agents for more focused and efficient downstream evaluation.

$$y_r = A_r(y_f, x_v, x_t, p_r, m), \quad \text{where } y_r \subseteq \mathcal{C} \tag{2}$$

Experts Group Building on the framework proposed by (Zhang et al., 2023b), we design an Expert Group A_g, composed of 8 specialized expert agents. The group is responsible for conducting fine-grained, multi-task analysis of memes (Task T3–T8), guided by the set of activated social categories y_r identified by the Router agent. We define the set of category-specific prompts as $\mathcal{P}_\mathcal{C} = \{p_c \mid c \in \mathcal{C}\}$, where each p_c corresponds to the prompt used by the expert agent assigned to analyze the meme from the perspective of social category c. For each activated social category $c \in y_r$,

Algorithm 1 AiO-MemeAgents

Input: A meme x consisting of image x_v and caption x_t; a multimodal LLM m; an agent set $\{A_f, A_r, A_g, A_o\}$ with corresponding prompts $\{p_f, p_r, \mathcal{P}_\mathcal{C}, p_o\}$
Output: A set of agent outputs Y
Initialize $Y \leftarrow \emptyset$, $y_g \leftarrow \emptyset$
// Figurative Agent
$y_f \leftarrow A_f(x_v, x_t, p_f, m)$
$y_r \leftarrow A_r(y_f, x_v, x_t, p_r, m)$ *// Router*
// Iterate over Experts activated by Router
for each category $c \in y_r$ **do**
  Fetch the mapped prompt p_c from $\mathcal{P}_\mathcal{C}$
  // Category-specific Expert
  $y_c \leftarrow A_g(y_f, x_v, x_t, p_c, m)$
  Add y_c to y_g
end for
// Orchestrator Agent
$y_o \leftarrow A_o(y_g, y_f, x_v, x_t, p_o, m)$
$Y \leftarrow \{y_f, y_r, y_g, y_o\}$
return Y

we retrieve the corresponding prompt p_c and activate the relevant expert A_g to perform downstream tasks T3 ~T8. This analysis leverages the shared figurative understanding y_f and the original meme inputs (x_v, x_t). The output of the expert agent for category c is denoted as:

$$y_c = A_g(y_f, x_v, x_t, p_c, m) \tag{3}$$

where m is the shared backbone model used across all experts. The experts group output is defined as:

$$y_g = \{y_c \mid c \in y_r\} \tag{4}$$

Orchestrator To finalize the process, we design an Orchestrator Agent A_o to summarize the out-

puts of the Figurative Explainer Agent and all activated Specialized Agents into a unified final report for tasks T1 ~T8. This agent consolidates multi-perspective analyses into an interpretable, task-specific summary of the meme.

$$y_o = A_o(y_g, y_f, x_v, x_t, p_o, m) \quad (5)$$

To streamline notation, we summarize the full procedure described in Equations 1 ~5 as a single function. We describe the technical details for our agentic framework in Algorithm 1. Specifically, we denote the zero-shot, multifaceted analysis of a meme $x = (x_v, x_t)$ performed by **AiO-MemeAgents** with backbone model m as:

$$\{y_f, y_r, y_g, y_o\} = \text{MemeAgents}(x, m) \quad (6)$$

3.2 Human-AI Annotation

Human-AI Annotation has attracted increasing interest in recent work (Zhang et al., 2025; Qian et al., 2025; Shen et al., 2025), where agentic generated responses is explored as a way to reduce reliance on costly manual labeling, and human verification can further improve annotation quality (Han et al., 2025). Our proposed Human-AI collaboration for annotation consists of two stages: 1) We introduce a Manager Agent A_{mgr} to consolidate two AiO-MemeAgents teams' outputs. The Manager Agent is powered by GPT-4o-mini as backbone model and guided by CoT prompting p_{mgr}. This agent integrates both teams' outputs with background knowledge (Tang et al., 2023) to produce high-quality multi-task annotations. We describe the details of automatic annotation with the Manager Agent in Algorithm 2 (Appendix B.3). 2) Human Verification for quality-control: domain experts review the synthesized outputs to attain high-quality labels, resolve inter-agent conflicts, discard low-confidence or unreliable predictions, and introduce new labels when existing annotations are insufficient. This human–AI collaborative design combines the scalability and consistency of agent-based automation with human judgment and domain awareness, ensuring high-quality and extensible annotations. The Human-AI collaboration produces a high-quality dataset, **Multifacet-Meme**, consisting of four subsets: D_f, D_r, D_g, and D_o, which correspond to the Figurative Agent, Router, Experts Group, and Orchestrator Agent, respectively.

3.3 Agentic Distillation

In this study, we use Multifacet-Meme to evaluate the zero-shot AiO-MemeAgents performance with different backbones. We further explore Multifacet-Meme for instruction fine-tuning, enabling student models to distill knowledge from the agentic framework to better handle the challenging multifaceted CMU. We follow the agentic distillation paradigm to transfer knowledge from our proposed multi-agent framework into compact student models. We fine-tune a student model (LLaMA3.2-11B-Vision) using LoRA-based instruction tuning on our Multifacet-Meme dataset (Figure 1 (3)), with the goal of replicating the reasoning patterns of AiO-MemeAgents. For each task-specific subset of Multifacet-Meme, we train a separate LoRA adapter and merge it with the frozen student model to obtain a lightweight model capable of replacing the corresponding task-specific agent. Finally, we aggregate the 4 fine-tuned lightweight models into a unified workflow, called **AiO-LLaMAs**, designed for efficient and scalable multifaceted CMU.

4 Experiments

4.1 Experimental Setup

Datasets We evaluate our approach on two public datasets, **HatReD** (Hee et al., 2023) and **MET-Meme** (Xu et al., 2022). The former focuses on harmful meme analysis, annotated with social category, social target, attack type, and reasoning. Additionally, we merge label information (social category and attack type) from WOAH-5 (Mathias et al., 2021), which shares overlapping memes with HatReD, originally sourced from the Hateful Memes Challenge (Kiela et al., 2020). The latter targets general-topic, multifaceted meme understanding with a focus on metaphor, sentiment, intention, and intensity. We randomly sample 3,300 memes from the two datasets (1,650 each), enabling a comprehensive evaluation of our work across both harmful and general-topic memes. Notably, each meme in our dataset is annotated across 8 distinct tasks, whereas prior studies typically employed only 2 or 4. This results in a total of 26,400 annotation instances, making our dataset the second largest by total task-sample pairs.

Baselines We reproduce MemeCLIP (Shah et al., 2024), a multi-label classification framework for meme understanding. We fine-tune task-specific MemeCLIP models for tasks T1–T6. We evaluate the zero-shot performance of AiO-MemeAgents using five different backbone models: GPT-4o-mini, Qwen2.5-VL-72B (Bai et al., 2023), Qwen2.5-VL-7B (Bai et al., 2023), LLaVA1.6-

vicuna-13B (Liu et al., 2023), and LLaMA3.2-11B-Vision (Grattafiori et al., 2024). This selection allows us to compare both closed- and open-source models, as well as large- and small-scale MLLMs. We use GPT-4o-mini as the backbone for the Manager Agent in the Human-AI Annotation stage (Section 3.2). We designate LLaMA3.2-11B-Vision as the student model for AiO-LLaMAs workflow distillation (Section 3.3) with 3-fold cross-validation strategy. Moreover, we run experiments with zero-shot and few-shot (3-shot) CoT paradigms with GPT-4o, Qwen-VL-Max, GPT-4o-mini, and Qwen2.5-VL-72B as backbones.

Evaluation We use BERTScore-F1 (Zhang et al., 2019) as the evaluation measure for generation tasks (T3 & T8), and standard F1-score for the remaining classification tasks. We provide more implementation details in Appendix B.4.

4.2 Main Results

In this study, we aim to answer the following 3 research questions:

RQ1: How effectively can AiO-MemeAgents perform Multifaceted CMU? We present the zero-shot performance of the proposed agent framework across 8 tasks with 5 backbone models in Table 3. 1) We observe that AiO-MemeAgents consistently outperform few-shot CoT baselines by a substantial margin across all eight tasks. For example, using Qwen2.5-VL-72B as the backbone model, AiO-MemeAgents achieve an average score of 71.16%, compared to 56.13% obtained with few-shot CoT on the HatReD dataset. Statistical significance tests further confirm that these improvements differences ($p < 0.05$). 2) Despite its best overall capabilities in few-shot CoT group, GPT-5.1 shows comparatively weaker performance in visual figurative reasoning than Qwen2.5-VL-72B and GPT-4o. Figurative language recognition remains a challenging task, consistently ranking among the top three hardest classification tasks across both datasets.

RQ2: To what extent can the distilled workflow AiO-LLaMAs replicate the performance of the full agentic framework's on each subtask? We visualize the performance among AiO-LLaMAs and baselines in Figure 2, with detailed results provided in Table 3. We observe that AiO-LLaMAs significantly outperform MemeAgents with small-scale MLLMs as backbones with $p < 0.001$. While there remains a performance gap between zero-shot AiO-MemeAgents with large-scale MLLMs like GPT-4o-mini and Qwen2.5-VL-72B, this is

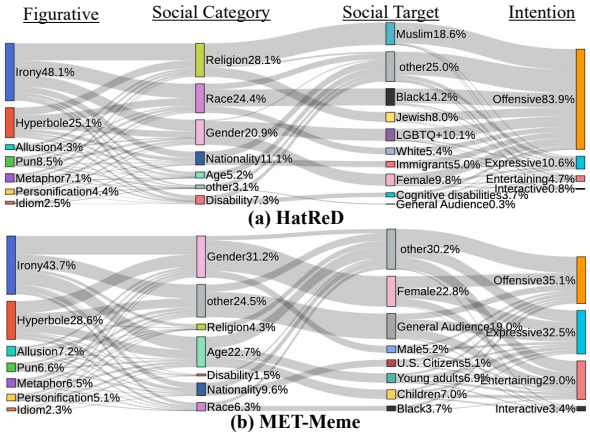


Figure 3: The distribution of **Multifacet-Meme** dataset across 4 dimensions: Figurative (T1), Social Category (T2), Social Target (T3), and Intention (T5).

expected given the substantial difference in parameter scale and training resources. Interestingly, AiO-LLaMAs performance closely matches few-shot CoT with advanced MLLMs such as GPT-4o, GPT-4o-mini, and Qwen2.5-VL-72B on HatReD, while an average gap of about 6% remains on MET-Meme. This indicates that general meme understanding is more challenging. Notably, AiO-LLaMAs are distilled from the automatically annotated Multifacet-Meme dataset, with only 1,650 samples per domain (3,300 in total). This result shows that Agentic Distillation can effectively infuse the knowledge of Aio-MemeAgent into the smaller student models with limited resources.

RQ3: How is figurative language used by meme creators in the context of multifaceted memes? To answer this question, we analyze the distribution of meme labels in **Multifacet-Meme** across 4 key tasks: Figurative Language (Liu et al., 2022), Social Category and Social Target (Hee et al., 2023), and Intention (Xu et al., 2022). These tasks collectively offer a unified view on meme understanding by integrating insights from prior multi-task work. The results are presented in Figure 3. We find that irony and hyperbole are the two dominant figurative expressions, appearing in over 72% of memes in HatReD and MET-Meme. Moreover, metaphor and personification occur in 11.5% of memes in HatReD and 11.6% in MET-Meme. This finding supports our hypothesis that a broader understanding of diverse figurative language is important for Multifaceted CMU. In contrast, prior studies have primarily focused on metaphor or considered only a small subset of figurative expressions (Hwang and Shwartz, 2023; Xu et al., 2024; Lin et al., 2024a;

	Backbone Model	Method	Figurative	Router	Orchestrator						Avg
			Figurative	*Social Category*	*Sentiment*	*Intention*	*Intensity*	*Attack Type*	*Social Target*†	*Explanation*†	
HatReD	GPT-5.1	3-shot CoT	29.41 (0.66)	**66.04** (1.02)	30.32 (1.02)	**82.25** (0.92)	**73.58** (1.09)	**40.50** (1.01)	85.57 (0.12)	88.85 (0.04)	**62.06**
	GPT-4o	3-shot CoT	35.96 (0.62)	56.12 (1.11)	22.48 (0.96)	74.46 (1.06)	68.91 (1.14)	34.01 (0.94)	87.62 (0.16)	89.28 (0.04)	58.60
	GPT-4o-mini	3-shot CoT	36.23 (0.63)	43.49 (1.14)	42.82 (1.11)	78.78 (0.99)	69.33 (1.14)	34.03 (0.94)	**87.80** (0.17)	**89.34** (0.05)	60.23
	Qwen2.5-VL-72B	3-shot CoT	**39.62** (0.54)	52.30 (1.13)	**50.43** (1.11)	68.67 (1.13)	33.45 (1.16)	29.73 (0.92)	86.76 (0.15)	88.09 (0.10)	56.13
	GPT-4o-mini	MemeAgents	45.19 (0.63)	56.04 (1.02)	**71.46** (0.97)	**74.17** (1.06)	**73.09** (1.09)	40.85 (1.01)	**92.32** (0.19)	**90.58** (0.08)	67.96
	Qwen2.5-VL-72B	MemeAgents	**78.46** (0.84)	**60.48** (1.05)	68.97 (0.99)	69.98 (1.11)	67.15 (1.16)	**43.36** (1.02)	92.07 (0.21)	88.82 (0.11)	**71.16**
	Qwen2.5-VL-7B	MemeAgents	32.76 (0.79)	41.51 (1.04)	32.00 (1.02)	56.03 (1.20)	22.55 (1.03)	24.52 (0.82)	88.67 (0.19)	88.15 (0.05)	48.27
	LLaVA1.6-vicuna-13B	MemeAgents	31.46 (0.72)	45.26 (0.97)	10.80 (0.70)	32.57 (1.14)	20.48 (0.99)	12.34 (0.66)	85.51 (0.16)	85.54 (0.09)	40.50
	LLaMA3.2-11B-Vision	MemeAgents	35.38 (0.61)	35.44 (1.01)	29.26 (1.02)	56.53 (1.21)	34.91 (1.17)	17.37 (0.73)	85.98 (0.16)	85.71 (0.09)	47.57
	MemeCLIP	Finetune	15.78 (0.83)	47.95 (1.05)	32.85 (0.82)	42.88 (1.21)	48.63 (1.23)	34.18 (0.76)	-	-	6.88
	AiO-LLaMAs (Ours)	Agentic Distillation	49.32 (0.80)	56.98 (0.94)	58.14 (0.98)	81.66 (0.92)	66.67 (1.16)	44.65 (0.88)	90.85 (0.20)	89.57 (0.05)	67.23
MET-Meme	GPT-5.1	3-shot CoT	37.33 (0.79)	**81.73** (0.92)	**46.78** (1.03)	51.63 (1.21)	**59.27** (1.21)	83.32 (0.89)	85.39 (0.10)	88.63 (0.03)	**66.76**
	GPT-4o	3-shot CoT	38.39 (0.74)	80.54 (0.95)	38.16 (1.01)	46.72 (1.21)	58.00 (1.22)	84.90 (0.85)	86.76 (0.14)	89.14 (0.04)	65.33
	GPT-4o-mini	3-shot CoT	33.33 (0.74)	80.22 (0.96)	38.13 (1.03)	**52.49** (1.21)	51.45 (1.23)	**85.45** (0.84)	**87.01** (0.14)	**89.41** (0.04)	64.69
	Qwen2.5-VL-72B	3-shot CoT	**41.81** (0.71)	80.16 (0.96)	43.34 (1.01)	48.64 (1.21)	32.91 (1.16)	76.61 (1.02)	85.92 (0.12)	88.23 (0.09)	62.20
	GPT-4o-mini	MemeAgents	53.92 (0.69)	**86.02** (0.82)	**55.53** (0.97)	54.14 (1.21)	**69.70** (1.13)	**89.35** (0.73)	**96.61** (0.14)	**91.48** (0.08)	**74.59**
	Qwen2.5-VL-72B	MemeAgents	**69.44** (0.86)	81.12 (0.93)	53.76 (0.96)	**54.57** (1.19)	63.27 (1.19)	87.97 (0.78)	92.24 (0.18)	89.53 (0.09)	73.99
	Qwen2.5-VL-7B	MemeAgents	33.74 (0.79)	69.06 (1.10)	16.76 (0.76)	40.96 (1.18)	37.94 (1.19)	74.92 (1.04)	87.51 (0.15)	88.12 (0.06)	56.13
	LLaVA1.6-vicuna-13B	MemeAgents	32.24 (0.74)	16.69 (0.82)	9.59 (0.62)	21.87 (1.00)	29.15 (1.12)	46.98 (1.22)	84.68 (0.13)	85.68 (0.09)	40.86
	LLaMA3.2-11B-Vision	MemeAgents	33.12 (0.71)	14.35 (0.80)	13.32 (0.74)	27.92 (1.09)	22.48 (1.03)	35.43 (1.17)	85.35 (0.13)	85.34 (0.09)	39.66
	MemeCLIP	Finetune	17.96 (0.84)	50.90 (1.01)	29.20 (0.74)	28.37 (1.18)	37.42 (0.83)	41.15 (0.13)	-	-	34.17
	AiO-LLaMAs (Ours)	Agentic Distillation	45.34 (0.80)	65.37 (1.11)	42.11 (0.92)	44.29 (1.19)	44.79 (1.22)	67.88 (1.11)	89.73 (0.17)	89.35 (0.05)	61.11

Table 3: Performance comparison among zero-shot AiO-MemeAgents, the distilled workflow AiO-LLaMAs, and 3-shot CoT. Default metric is F1-score (%). † Social Target & Explanation uses BERTScore-F1 (%) due to its emphasis on semantic similarity.

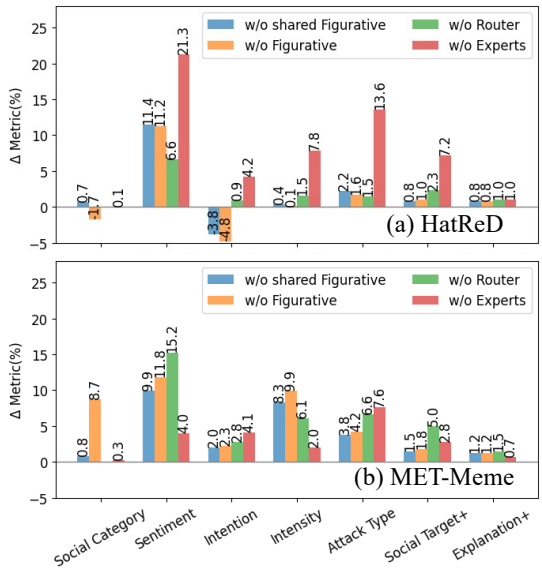
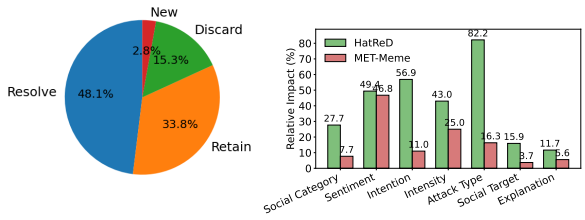


Figure 4: Ablation study results with Qwen2.5-VL-72B as the backbone for Aio-MemeAgent. Each bar shows the performance drop (Δ Metric).



(a) Human verification (b) Error propagation analysis.

Figure 5: Human verification and error propagation analysis in the MemeAgent framework.

Chauhan et al., 2020). Our study addresses this gap by exploring a richer taxonomy of 8 figurative expressions across both harmful and general memes. For further analysis, refer to Appendix B.1.

4.3 Analysis

Human Verification for Data Quality Control Synthetic datasets generated by agentic frameworks may contain low-quality or conflicting annotations that are misaligned with human preferences. To address this issue, we propose a *Human–AI annotation* paradigm that incorporates human supervision after the Manager Agent aggregates final outputs from two MemeAgent teams. This design enables effective quality control by combining automated reasoning with human judgment. As illustrated in Figure 5a, among 26,400 task–sample pairs produced by the Manager Agent, 33.8% of the labels are identified as high-quality and directly retained. In addition, 48.1% of the synthetic labels require domain experts to resolve annotation conflicts, 15.3% are discarded due to low quality, and 2.8% correspond to newly introduced labels. To further evaluate the quality of **Multifacet-Meme**, we recruited four English-proficient graduate students as annotators. The resulting dataset demonstrates high annotation quality, achieving an average human satisfaction rate of 89.87%.

Evaluation for Error Propagation. We conduct a *forced-setting* experiment in which downstream tasks are provided with ground-truth outputs from the Figurative Agent (i.e., assuming 100% figura-

tive understanding accuracy), using Qwen2.5-VL-72B as the backbone model. We define the *Relative Impact Ratio* as:

$$\textbf{Relative Impact} = \frac{\Delta_{\text{sub-task}}}{\Delta_{\text{Figurative}}} \tag{7}$$

where Δ denotes the change in task performance shown in Table 3 under the *forced-setting* condition. The Relative Impact Ratio quantifies whether errors originating from figurative understanding are amplified or attenuated along the agent collaboration pipeline. As shown in Figure 5b, Relative Impact ratios below 1 across tasks and domains indicate that error propagation is generally limited under the proposed **MemeAgent** framework. The largest relative impact ratio is 82.2% for the Attack Type Classification task on the HatReD dataset, suggesting that accurate figurative understanding can contributes to identifying attack types. This observation aligns with human intuition, as recognizing underlying figurative cues is often critical for correctly interpreting the intent and nature of harmful memes.

Ablation Study. We conduct an ablation study to assess the contribution of each component in AiO-MemeAgents, using Qwen2.5-VL-72B as the backbone. We evaluate the following variants: (a) *w/o* shared Figurative, which removes the shared output from the Figurative Agent; (b) *w/o* Figurative, which removes the Figurative Agent entirely; (c) *w/o* Router, which forwards all memes to all Expert agents; and (d) *w/o* Experts, which removes the Experts Group. Results are shown in Figure 4.

The impact of individual components varies across datasets. On HatReD, the Experts Group contributes broadly, yielding gains in Sentiment Classification (+21.3%) and Attack Type Detection (+13.6%). On MET-Meme, different components dominate different tasks: the Router contributes most to Sentiment (+15.2%), while the Figurative Agent provides the largest gain in Intensity (+9.9%). Interestingly, removing the Figurative Agent or its shared signals improves performance on Social Category and Intention in HatReD, but this effect does not generalize to MET-Meme, where these components instead improve Social Category, Sentiment, and Intensity. This difference may stem from HatReD memes relying more on direct expressions of intent toward social groups. Overall, integrating all components in **AiO-MemeAgents** supports robust generalization across harmful and general meme datasets, leading to strong performance on Multifaceted CMU.

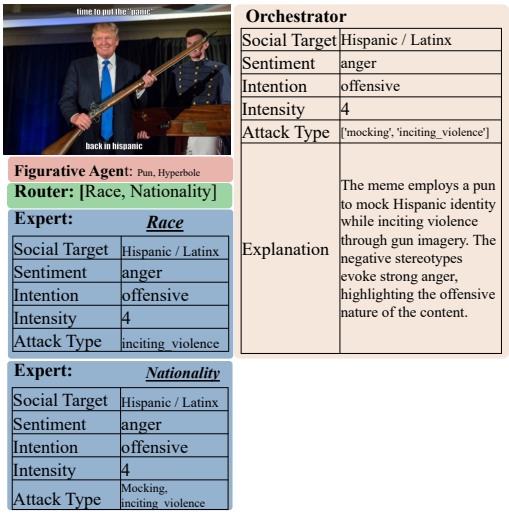


Figure 6: Example from HatReD with caption 'time to put the "panic" back in hispanic'.

Case Study To provide a intuitive illustration of our agentic framework, Figure 6 presents an representative examples. The Figurative Agent jointly analyzes the meme's text and image, identifying hyperbole—the exaggerated violent imagery combined with the pun "put the panic back in Hispanic"—as the core figurative device. Guided by this interpretation, the Router directs the meme to Race and Nationality experts. The Orchestrator then integrates these multimodal cues into an interpretable explanation, demonstrating how hyperbolic and pun-based visual and textual elements jointly convey the meme's hateful message.

5 Conclusion & Future Work

In this work, we provided a comprehensive analysis of the multifaceted characteristics of memes, highlighting the important role of figurative language in improving multi-task meme understanding. We introduced **AiO-MemeAgents**, an All-in-One multi-agent framework for Multifaceted CMU. We also curated **Multifacet-Meme**, a high-quality, automatically annotated dataset with agents collaboration. We distilled a high performance workflow with small-scale student models, **AiO-LLaMAs**. Experiments demonstrated the effectiveness and generalizability of our work, and show that agentic distillation can effectively empower student models to replicate the reasoning patterns of AiO-MemeAgents.

6 Limitations

While our proposed pipeline shows high performance and effectiveness in Multifaceted CMU, several limitations remain. First, although we cover 8 distinct tasks of meme understanding, there are other tasks yet to be explored. For example, integrating persuasive techniques (Hasanain et al., 2023; Dimitrov et al., 2024) detection could further enrich the capability of our framework. Additionally, the ambiguity of multimodal figurative expressions in memes increases annotation difficulty (Ziegler et al., 2020). Future work can explore strategies to address this issue.

Ethical Considerations

This work studies computational meme understanding using publicly available meme datasets and online content released for research purposes. The proposed framework focuses on analyzing multimodal signals (images and text) to support tasks related to figurative language, emotion, and social harm, and does not involve interaction with human subjects or experiments on animals.

Human involvement in this work is limited to a low-risk evaluation and validation setting. Specifically, graduate student annotators assess the quality of automatically generated labels and agent outputs using binary judgments. Annotators are not asked to produce original labels, disclose personal information, or engage with private or sensitive data. All annotators participated voluntarily and evaluated publicly available meme content.

Because memes may contain offensive, hateful, or sensitive material, we acknowledge the potential psychological discomfort associated with exposure to such content. Annotators were informed of this risk in advance and were allowed to discontinue participation at any time. We do not collect or release personally identifiable information about individuals depicted in the memes, and we restrict dataset usage to research purposes.

Our dataset construction relies on a Human–AI collaborative annotation process, combining agent-generated outputs with human verification. To mitigate risks associated with incorrect, biased, or misleading model outputs, we employ multi-agent cross-checking, background knowledge constraints, and human evaluation to improve label quality and transparency. We emphasize that the annotations reflect interpretations of meme content rather than ground-truth intent, particularly for figurative and ambiguous cases.

We also recognize that multi-agent LLM-based pipelines incur nontrivial computational cost. To promote responsible use, we introduce an agentic distillation workflow that transfers reasoning capabilities into smaller, more efficient student models, reducing inference-time resource consumption and supporting more sustainable deployment.

Finally, the methods and datasets introduced in this work are intended for research and benchmarking purposes. While our framework can assist in analyzing harmful content, it is not designed as a standalone moderation or enforcement tool. We declare no conflicts of interest.

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A Prompts Details

A.1 Prompt engineering strategies

The related studies that informed our prompt design are as follows: 1) Figurative agent: Xu et al. (2022) inspired our prompt for visual metaphor analysis,

and Liu et al. (2022) encouraged us to further extend the scope to general figurative expressions. 2) Router: DeepSeek-AI et al. (2024) inspired us to incorporate a router agent to avoid unnecessary computation. 3) Experts Group: Hee et al. (2023) motivated us to create a group of agents representing different social categories for meme analysis.

A.2 Figurative Agent

You are an expert at analysis and explanation of the figurative expressions used in Memes. ROLE: Identify figurative expressions in the meme text/description (at most 2 types). Optional figurative categories are: {Figurative Language Taxonomy}

A.3 Router

You are a Router responsible for selecting which aspects should be censored to better understand a meme's sentiment, intention, intensity, attack type, and target group.
Task: Given a meme (text + image) and figurative language expert explanation for the meme, determine which of the following aspects should be censored to enhance interpretability: Social Category list Your decision should be based on: 1) Target Relevance. Does the meme explicitly or implicitly reference any of these aspects? 2) Sentiment & Intensity Impact. Does censoring a particular aspect help better analyze how strong the sentiment or attack is? 3) Contextual Necessity. Is the aspect central to understanding the meme?

A.4 Expert Agents

You are the Activated Social Category Agent. **ROLE**: Classify how this meme addresses or affects individuals/groups based on race or ethnicity. **EXAMPLE GROUPS** Example list of Social Target options

TASK REQUIREMENTS 1) Determine sentiment, intention, intensity, attack type, and target group if relevant. a) Sentiment: Identify the emotion that the creator intends to evoke in the audience, options [happiness, love, anger, sorrow, fear, hate, surprise, neutral] b) Intention: Determine the purpose or motivation behind creating the meme, options [interactive, expressive, purely entertaining, offensive, other] c) Intensity: Assess the strength or degree of the sentiment being conveyed. A numerical score from 1 to 4, where 1 = very mild or weak, 4 = very strong or intense. d) Target Group: Identify the intended audience demographic the meme is created for, options Social target options e) Attack Type: If the meme contains harmful or offensive content, specify the type of attack being used, options [dehumanizing, inferiority, inciting_violence, mocking, contempt, slurs, exclusion, none] 2) If no race-based target, "None". 3) Keep reasoning concise and within 30 words. 4) The Attack Type category applies exclusively to hateful or offensive memes; in all other cases, it should be marked as "None."

A.5 Orchestrator Agent

You are the Aggregator/Orchestrator. **ROLE:** Combine outputs from the Metaphor Explanation Agent and each Specialized Agent into a final report. Multifaceted CMU definition
TASK REQUIREMENTS: 1) Produce a single JSON report that includes: a) An aggregated conclusion: b)possibly identify any dominant target group(s) c) overall sentiment, intention, intensity, and attack type (if relevant) d)note that memes can serve multiple purposes or intentions (not always hateful) 2) Keep the final reasoning concise. 3) The Attack Type category applies exclusively to hateful or offensive memes; in all other cases, it should be marked as "None."

A.6 Manager Agent

You are a manager agent. Based on the task guidelines, the meme's golden labels information, and results from sub-agents, generate the final output. Ensure your output aligns with the golden label information (excluding the figurative expressions section), and utilize the results from the sub-agents to generate the final output. Task guideline: Agents prompt Input Meme captions: Caption. Meme golden labels information: Background information. Result from sub-agents: outputs from other AiO-MemeAgents teams.

A.7 CoT

Instructions: Given a meme (image and caption), answer each of the following 8 questions with a single, clear choice or response. Output your results as a JSON object.
CoT Questions (One per Task):

T1 (Figurative): Which types (up to two) of figurative language is most prominent in the meme? Options: "metaphor", "irony", "hyperbole", "onomatopoeia", "personification", "allusion", "idiom", "pun", "none"

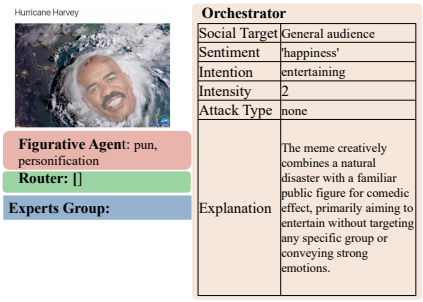

Figure 7: Example from MET-Meme with caption "Hurricane Harvey". Router did not activate any Experts, so there is no need for further analysis by Experts Group.

T2 (Social Category): Which social category is primarily referenced or targeted by the meme? Options: "gender", "race", "age", "religion", "disability", "nationality", "none"

.....

T7 (Attack Type): If the meme is offensive or harmful, what are the primary attack types (up to two)? Options: "dehumanizing", "inferiority", "inciting violence", "mocking", "contempt", "slurs", "exclusion", "none"

T8 (Explanation): Briefly explain your choices above (1–2 sentences).

Note: For each task with a list of options, please ensure your answer is selected from that list.

A.8 Case Study

In Figure 7, the Figurative Agent identifies pun and personification—combining "Hurricane Harvey" with a celebrity's image to create a lighthearted, figurative joke. Since no hateful or identity-targeting cues are detected, the Router remains inactive and does not activate any Expert Agents. This selective routing mechanism avoids unnecessary expert-level analysis, conserving computational resources while maintaining interpretability. The Orchestrator then concludes the meme's entertaining intent, low intensity, and no attack type, generating an explanation that highlights benign humor rather than hostility.

B Methodology Details

B.1 Figurative Language Taxonomy

We consider the following figurative language types according to the knowledge from Wikipedia page: https://en.wikipedia.org/wiki/Literal_and_figurative_language.

- Metaphor: A comparison between two unlike things. e.g., "Time is a thief."
- Onomatopoeia: Words that imitate natural sounds. e.g., "buzz," "crash," "sizzle," "bang."
- Personification: Giving human traits to non-human things or ideas. e.g., "The wind whispered through the trees."
- Irony: Irony is when there's a contrast between expectation and reality — what is said or expected is different from (or opposite to) what actually happens or is meant.
- Hyperbole: An exaggeration for emphasis or effect. e.g., "I'm so hungry I could eat a horse."
- Allusion: A reference to a well-known person, place, event, or work of art/literature. e.g., He met his Waterloo." (Alludes to Napoleon's defeat).
- Idiom: A common phrase whose meaning is different from its literal meaning. e.g., "Kick the bucket" = to die; "Break the ice" = to start a conversation.
- Pun: A play on words, often using words that sound alike or have multiple meanings. e.g., "Time flies like an arrow; fruit flies like a banana."

B.2 Social Target Taxonomy

The Social Target taxonomy supports nuanced detection of social entity specific harms. We list a few examples as the target Entities for each social category, and prompt the agent to either generate the most proper fine-grained target group or choose the most suitable one from the pre-defined list of each social category:

- **Gender**: Male, Female, LGBTQ+.
- **Race**: Black, White, Hispanic, Asian, Indigenous, etc.
- **Religion**: Christian, Muslim, Jewish, etc.
- **Disability**: Physical, sensory, cognitive, neurodivergent, etc.
- **Age**: Children, Teenagers, Young Adults, Middle-Aged Adults, Older Adults, etc.
- **Nationality**: Citizens, Immigrants, Refugees, Visitors, etc.

Algorithm 2 Manager Agent

Input: A meme dataset D, A Manager Agent A_{mgr}, AiO-MemeAgents and it's prompts set $\{p_f, p_r, \mathcal{P}_\mathcal{C}, p_o\}$ and 2 backbones m^a, m^b
Output: Multifacet-Meme Dataset $\{D_f, D_r, D_g, D_o\}$
Initialize $\{D_f, D_r, D_g, D_o\} \leftarrow \{\emptyset, \emptyset, \emptyset, \emptyset\}$
for each meme $x \in D$ **do**
  // zero-shot AiO-MemeAgents (Eq. 6)
  $\{y_f^a, y_r^a, y_g^a, y_o^a\} \leftarrow \text{MemeAgents}(x, m^a)$
  $\{y_f^b, y_r^b, y_g^b, y_o^b\} \leftarrow \text{MemeAgents}(x, m^b)$
  Get background information of x as k
  // Annotate Figurative Agent response
  $y_f \leftarrow A_{mgr}(y_f^a, y_f^b, x, k, p_f)$
  // Figurative response dataset
  Add (x, p_f, y_f) to D_f
  // Annotate Router response
  $y_r \leftarrow A_{mgr}(y_f, y_r^a, y_r^b, x, k, p_r)$
  Add (x, p_r, y_r) to D_r
  // Iterate over activated Experts
  for each category $c \in y_r$ **do**
    Fetch the mapped prompt p_c from $\mathcal{P}_\mathcal{C}$
    // Annotate Expert response
    $y_c \leftarrow A_{mgr}(y_f, y_g^a, y_g^b, x, k, p_c)$
    Add (x, p_c, y_c) to D_g
    Add y_c to y_g
  end for
  // Annotate Orchestrator response
  $y_o \leftarrow A_{mgr}(y_f, y_g, y_o^a, y_o^b, x, k, p_o)$
  Add (x, p_o, y_o) to D_o
end for
return $\{D_f, D_r, D_g, D_o\}$

B.3 Manager Agent details

We present algorithm for Manager Agent in Alg 2. Existing datasets cover only subsets of the label space required for Multifaceted CMU (Table 1), we treat available labels as background knowledge k to guide the annotation of missing tasks. (Tang et al., 2023)

B.4 Implementation Details

We use the official APIs to access the large-scale MLLMs, GPT-4o-mini and Qwen2.5-VL-72B. The 3 small-scale MLLMs are hosted locally, with model parameters obtained from Hugging Face (Wolf et al., 2020). We set the maximum output token limit to 400 and the temperature to 0.85 for all MLLMs. For Agentic Distillation, we fine-tune the student models using the Unsloth (Daniel Han and team, 2023) package with 16-bit LoRA and set rank size to 16. We adopt a 3-fold strategy and set epochs to 3. Other hyperparameters set with default values. All experiments are conducted on a Linux server with 3 NVIDIA RTX A6000 GPUs.

B.5 Strategies to address the ambiguity of figurative expressions

The focus of this work is the agentic framework and agentic distillation for multifaceted meme understanding, not on an in-depth study of ambiguity inherent in figurative language. While there are already many works (Chakrabarty et al., 2022; Kim et al., 2024) addressing this topic, we acknowledge that it remains an open problem and have highlighted it as an important limitation and a direction for future work.